Knoelle C. Grassi

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SUMMARY

Analyst with a B.S. in Applied Mathematics and hands-on experience developing optimization models, statistical analyses, and forecasting solutions for complex operational problems. Proficient in Python, SQL, Tableau, and Power BI for data modeling, exploratory analysis, and decision support. Experienced in mathematical optimization, probability modeling, and predictive analytics across workforce allocation, crime forecasting, and operational planning projects. Strong foundation in nonlinear systems, optimization, and statistical inference.

EXPERIENCES

Spirit Halloween

Sales Associate, October 2025 – November 2025

- Increased average transaction value by an estimated 10–15% through targeted product recommendations and add-on sales
- Assisted 50+ customers per shift, maintaining high satisfaction during peak seasonal volume
- Supported daily merchandising resets, reducing item misplacement

The Whiting-Turner Contracting Company

Intern, May 2025 – August 2025

- Built automated KPI reports using SQL, Excel, and Power BI, reducing manual reporting time by ~40%
- Designed interactive dashboards consumed by senior leadership to track equipment work orders

Rick's Ranchwear, Inc.

Receiving Clerk, June 2021 – August 2024

- Created Excel-based reports tracking top-selling products, inventory turnover, and logistics metrics for management
- Supported logistics operations processing Amazon orders daily, improving order accuracy and turnaround
- Collaborated directly with the VP and logistics leadership to support inventory planning and reporting needs

Tata Data Visualization Job Simulation (Forage)

December 2025

- Analyzed client business data and created executive-ready visualizations aligned with stakeholder decision needs
- Developed structured analytical questions for senior leadership meetings
- Translated raw data into clear insights using professional data storytelling technique

SKILLS

- **Programming & Tools:** Python (Pandas, NumPy, PuLP), SQL (MySQL), R, MATLAB, Tableau, Power BI, Excel, Jupyter, Git
- **Analytics & Modeling:** Data cleaning, ETL pipelines, statistical analysis, probability modeling, forecasting, optimization, algorithm development, dashboarding
- **Mathematics & Theory:** Applied mathematics, nonlinear systems, bifurcation analysis, numerical methods, optimization theory, Fourier analysis

EDUCATION

Florida Polytechnic University, Lakeland, Florida

Bachelor of Science in Applied Mathematics (August 2025)

PROJECTS

311 Workforce Optimization Model *(December* 2025 – *April* 2026)

- Formulated and solved constrained workforce allocation problems using historical NYC 311 service data to optimize staffing distribution across boroughs
- Implemented linear programming models in Python (PuLP) to minimize proportional mismatch between service workload and allocated personnel under operational constraints
- Evaluated alternative allocation frameworks including forecast-adjusted, median-based, and heuristic staffing models to compare decision-making strategies
- Conducted sensitivity analysis on model parameters and constraints to quantify allocation stability under varying operational assumptions

Medical Appointment No-Show Analysis *(November* 2025 – *December* 2025)

- Analyzed 100k+ appointment records using Python to identify drivers of no-show risk
- Built a risk-scoring model combining demographic, scheduling, and condition-based features
- Designed 4 Tableau dashboards to support operational decision-making and workflow optimization
- Quantified the impact of key variables, identifying factors contributing to up to 20% increase in no-show probability

Predictive Crime Analysis Model (Capstone Project) *(August* 2024 – *May* 2025)

- Modeled multi-year police incident data to estimate crime likelihood by time and location
- Developed probability algorithms for event frequency and category prediction
- Integrated AI-generated predictions and evaluated performance using accuracy and error metrics

Wildlife Data Warehouse & Tableau Insights (May 2025)

- Designed a star-schema data warehouse integrating species, environmental, trade, and geographic datasets
- Processed and modeled thousands of spatiotemporal observations
- Built Tableau dashboards identifying high-risk ecological regions for conservation focus

Rössler System Modeling & Bifurcation Analysis *(November 2024)*

- Modeled nonlinear chaotic systems and identified critical bifurcation parameters via numerical simulation
- Validated analytical results against computational solutions in MATLAB
- Produced reproducible plots and simulations for academic reporting

Dirichlet Boundary Value Problem Solver (November 2024)

- Solved multi-region Dirichlet problems using harmonic functions and conformal mapping
- Derived analytic solutions via complex logarithms and argument functions
- Visualized isotherms and heat flux using MATLAB and Python

Game Analytics Database System (November 2024)

- Designed and implemented a relational database supporting user activity tracking and quiz performance analytics
- Developed SQL queries for score tracking, usage patterns, and trend reporting

The Fourier Analysis Final Project (November 2024)

- Applied FFT-based high-pass filtering to remove low-frequency noise from audio signals
- Analyzed signal behavior in time and frequency domains
- Optimized filter length via coefficient thresholding